Merge Overlapping Intervals

AlgoExpert · Medium · Arrays

The problem

- Merge all overlapping [start, end] intervals; result sorted by start.

Intuition

- After sorting by start, overlapping intervals are adjacent, so one pass can merge them.

Brute force vs optimal

- Brute force: repeatedly merge any overlapping pair → $O(n^2)$ or worse
- Optimal: sort by start, single sweep → O(n log n)
- After sorting, only the last merged interval can overlap

Approach

- 1. Sort by start
- 2. Keep the current merged interval
- 3. Extend its end on overlap
- 4. Otherwise start a new one

Worked example

sorted intervals



merged



[1,2] no overlap → start new interval

Worked example

sorted intervals



merged



[3,5] no overlap → start new interval

Worked example

sorted intervals



merged



[4,7] overlaps → extend to [3, 7]

Worked example

sorted intervals



merged



[6,8] overlaps → extend to [3, 8]

Worked example

sorted intervals



merged



[9,10] no overlap → start new interval

Java solution

```java
public class MergeOverlappingIntervals {
    public static int[][] mergeOverlappingIntervals(int[][] intervals) {
        int[][] s = intervals.clone();
        Arrays.sort(s, (x, y) -> Integer.compare(x[0], y[0]));
        List<int[]> out = new ArrayList<>();
        for (int[] iv : s) {
            if (!out.isEmpty() && out.get(out.size() - 1)[1] >= iv[0])
                out.get(out.size() - 1)[1] = Math.max(out.get(out.size() - 1)[1], iv[1]);
            else out.add(new int[]{iv[0], iv[1]});
        }
        return out.toArray(new int[0][]);
    }
}
```

Complexity

- Time: O(n log n)
- Space: O(n)

Edge cases & pitfalls

- Touching intervals [3,5] and [5,7] overlap (≥)
- Fully contained intervals: take max of ends
- Don't mutate the caller's array

Interview tips & follow-ups

- Follow-up: insert a new interval into a sorted list
- Follow-up: minimum meeting rooms (Laptop Rentals)

Recap

- After sorting by start, overlapping intervals are adjacent, so one pass can merge them.
- Complexity: O(n log n)